

CHAPTER 3: HOW SECURITIES ARE TRADED

8. a. The buy order will be filled at the best limit-sell order price: \$50.25
- b. The next market buy order will be filled at the next-best limit-sell order price: \$51.50
- c. You would want to increase your inventory. There is considerable buying demand at prices just below \$50, indicating that downside risk is limited. In contrast, limit sell orders are sparse, indicating that a moderate buy order could result in a substantial price increase.

11. The total cost of the purchase is: $\$20 \times 1,000 = \$20,000$

You borrow \$5,000 from your broker and invest \$15,000 of your own funds. Your margin account starts out with equity of \$15,000.

- a. (i) Equity increases to: $(\$22 \times 1,000) - \$5,000 = \$17,000$
Percentage gain = $\$2,000/\$15,000 = 0.1333$, or 13.33%
- (ii) With price unchanged, equity is unchanged.
Percentage gain = zero
- (iii) Equity falls to $(\$18 \times 1,000) - \$5,000 = \$13,000$
Percentage gain = $(-\$2,000/\$15,000) = -0.1333$, or -13.33%

The relationship between the percentage return and the percentage change in the price of the stock is given by:

$$\% \text{ return} = \% \text{ change in price} \times \frac{\text{Total investment}}{\text{Investor's initial equity}} = \% \text{ change in price} \times 1.333$$

For example, when the stock price rises from \$20 to \$22, the percentage change in price is 10%, while the percentage gain for the investor is:

$$\% \text{ return} = 10\% \times \frac{\$20,000}{\$15,000} = 13.33\%$$

- b. The value of the 1,000 shares is $1,000P$. Equity is $(1,000P - \$5,000)$. You will receive a margin call when:

$$\frac{1,000P - \$5,000}{1,000P} = 0.25 \Rightarrow \text{when } P = \$6.67 \text{ or lower}$$

- c. The value of the 1,000 shares is $1,000P$. But now you have borrowed \$10,000 instead of \$5,000. Therefore, equity is $(1,000P - \$10,000)$. You will receive a margin call when:

$$\frac{1,000P - \$10,000}{1,000P} = 0.25 \Rightarrow \text{when } P = \$13.33 \text{ or lower}$$

With less equity in the account, you are far more vulnerable to a margin call.

- b. By the end of the year, the amount of the loan owed to the broker grows to:

$$\$5,000 \times 1.08 = \$5,400$$

The equity in your account is $(1,000P - \$5,400)$. Initial equity was \$15,000. Therefore, your rate of return after one year is as follows:

$$(i) \quad \frac{(1,000 \times \$22) - \$5,400 - \$15,000}{\$15,000} = 0.1067, \text{ or } 10.67\%$$

$$(ii) \quad \frac{(1,000 \times \$20) - \$5,400 - \$15,000}{\$15,000} = -0.0267, \text{ or } -2.67\%$$

$$(iii) \quad \frac{(1,000 \times \$18) - \$5,400 - \$15,000}{\$15,000} = -0.1600, \text{ or } -16.00\%$$

The relationship between the percentage return and the percentage change in the price of Xtel is given by:

$$\% \text{ return} = \left(\% \text{ change in price} \times \frac{\text{Total investment}}{\text{Investor's initial equity}} \right) - \left(8\% \times \frac{\text{Funds borrowed}}{\text{Investor's initial equity}} \right)$$

For example, when the stock price rises from \$40 to \$44, the percentage change in price is 10%, while the percentage gain for the investor is:

$$\left(10\% \times \frac{\$20,000}{\$15,000} \right) - \left(8\% \times \frac{\$5,000}{\$15,000} \right) = 10.67\%$$

- e. The value of the 1000 shares is $1,000P$. Equity is $(1,000P - \$5,400)$. You will receive a margin call when:

$$\frac{1,000P - \$5,400}{1,000P} = 0.25 \Rightarrow \text{when } P = \$7.20 \text{ or lower}$$

12. a. The gain or loss on the short position is: $(-1,000 \times \Delta P)$
Invested funds = \$15,000

Therefore: rate of return = $(-1,000 \times \Delta P)/15,000$

The rate of return in each of the three scenarios is:

- (i) Rate of return = $(-1,000 \times \$2)/\$15,000 = -0.1333$, or -13.33%
- (ii) Rate of return = $(-1,000 \times \$0)/\$15,000 = 0\%$
- (iii) Rate of return = $[-1,000 \times (-\$2)]/\$15,000 = +0.1333$, or $+13.33\%$

b. Total assets in the margin account equal:

$\$20,000$ (from the sale of the stock) + $\$15,000$ (the initial margin) = $\$35,000$

Liabilities are $500P$. You will receive a margin call when:

$$\frac{\$35,000 - 1,000P}{1,000P} = 0.25 \Rightarrow \text{when } P = \$28 \text{ or higher}$$

c. With a $\$1$ dividend, the short position must now pay on the borrowed shares:
 $(\$1/\text{share} \times 1000 \text{ shares}) = \1000 . Rate of return is now:

$$[(-1,000 \times \Delta P) - 1,000]/15,000$$

- (i) Rate of return = $[(-1,000 \times \$2) - \$1,000]/\$15,000 = -0.2000$, or -20.00%
- (ii) Rate of return = $[(-1,000 \times \$0) - \$1,000]/\$15,000 = -0.0667$, or -6.67%
- (iii) Rate of return = $[(-1,000) \times (-\$2) - \$1,000]/\$15,000 = +0.067$, or $+6.67\%$

Total assets are $\$35,000$, and liabilities are $(1,000P + 1,000)$. A margin call will be issued when:

$$\frac{35,000 - 1,000P - 1,000}{1,000P} = 0.25 \Rightarrow \text{when } P = \$27.2 \text{ or higher}$$

14. a. $\$55.50$
- b. $\$55.25$
- c. The trade will not be executed because the bid price is lower than the price specified in the limit-sell order.
- d. The trade will not be executed because the asked price is greater than the price specified in the limit-buy order.

CFA PROBLEMS

1.
 - a. In addition to the explicit fees of \$70,000, FBN appears to have paid an implicit price in underpricing of the IPO. The underpricing is \$3 per share, or a total of \$300,000, implying total costs of \$370,000.
 - b. No. The underwriters do not capture the part of the costs corresponding to the underpricing. The underpricing may be a rational marketing strategy. Without it, the underwriters would need to spend more resources in order to place the issue with the public. The underwriters would then need to charge higher explicit fees to the issuing firm. The issuing firm may be just as well off paying the implicit issuance cost represented by the underpricing.
2. (d) The broker will sell, at current market price, after the first transaction at \$55 or less, so it is uncertain as to final price per share as the price drops.